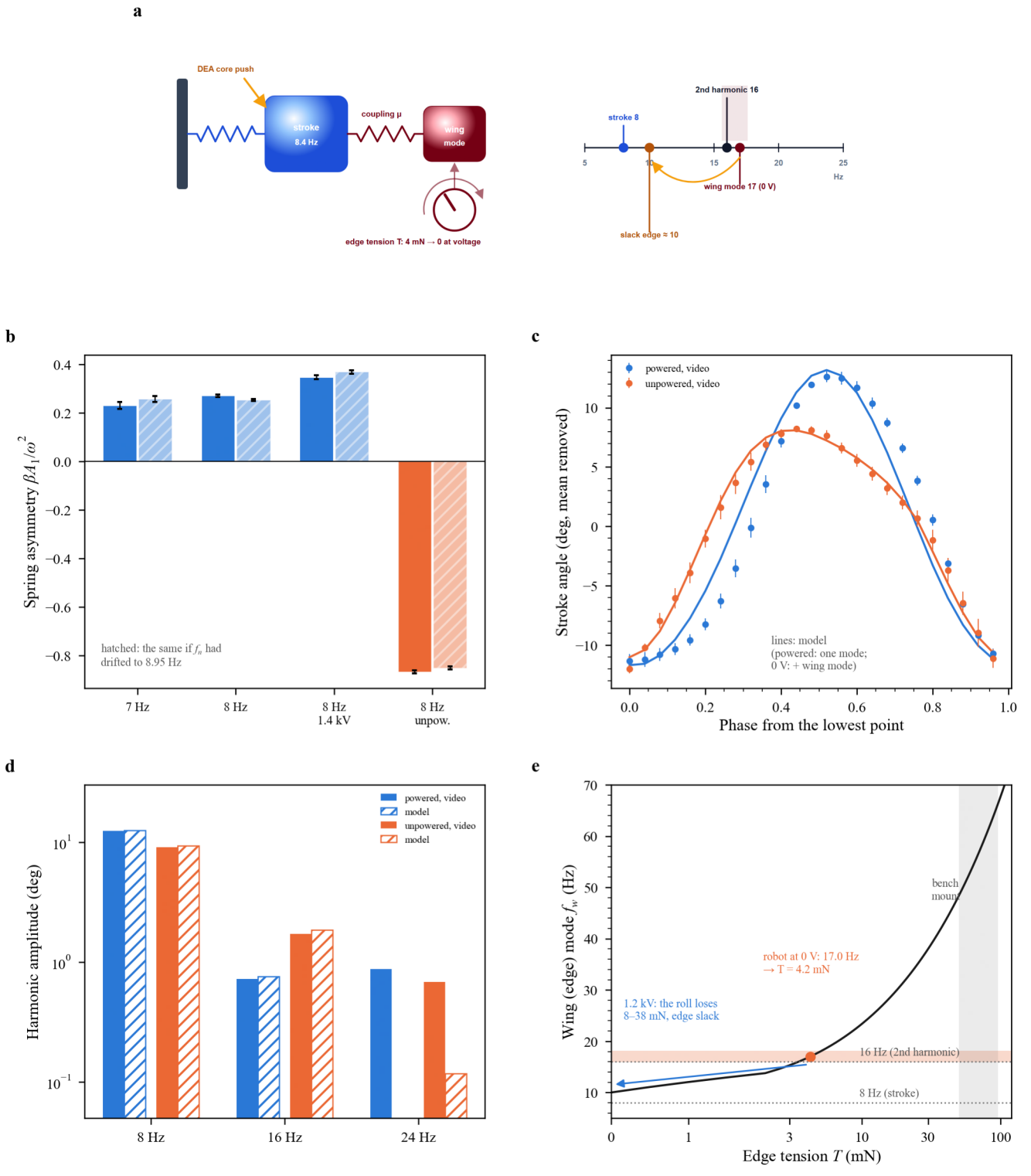

a
DEA core push
stroke
8.4 Hz
coupling μ
wing
mode
edge tension T: 4 mN → 0 at voltage
2nd harmonic 16
stroke 8
5
10
15
20
25
Hz
wing mode 17 (0 V)
slack edge ≈ 10
b
Spring asymmetry $\beta A_1/\omega^2$
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
hatched: the same if f_n had
drifted to 8.95 Hz
7 Hz
8 Hz
8 Hz
1.4 kV
8 Hz
unpow.
c
powered, video
unpowered, video
Stroke angle (deg, mean removed)
10
5
0
−5
−10
lines: model
(powered: one mode;
0 V: + wing mode)
0.0
0.2
0.4
0.6
0.8
1.0
Phase from the lowest point
d
powered, video
model
unpowered, video
model
Harmonic amplitude (deg)
10^1
10^0
10^{-1}
8 Hz
16 Hz
24 Hz
e
Wing (edge) mode f_w (Hz)
70
60
50
40
30
20
10
bench
mount
robot at 0 V: 17.0 Hz
→ T = 4.2 mN
1.2 kV: the roll loses
8–38 mN, edge slack
16 Hz (2nd harmonic)
8 Hz (stroke)
0
1
3
10
30
100
Edge tension T (mN)